Project Title

Competition Track

Track A : Technology & Engineering Solutions

**Main Challenge Area**
**Smart Water & Digital Innovation**

Problem Statement
Briefly describe the water-related problem that your project seeks to address (Max 250 words)

**When a pipe bursts, a treatment plant shuts down because of river pollution, or scheduled maintenance cuts supply, utilities in Selangor deploy water tankers and static tanks to affected communities. Tankers, water and time are limited, so every dispatch decision determines who receives water and who waits. These decisions are often guided by the order in which complaints arrive or by which site is easiest to reach. Such rules are simple, but they ignore how much water each community still has, how many people depend on it and how quickly it will run out. As a result, some communities run dry while others still have usable reserves.**

**Delivery to a location is also not the same as access. In high-rise flats, older residents, persons with disabilities and households without help may be unable to carry water from a static tank to their homes.**

**Access to safe water is a recognised human right, and the Sphere humanitarian standard sets a minimum of 15 litres per person per day in emergencies. This project asks: under the same tanker and water constraints, can a transparent allocation method reduce unmet essential water needs compared with first-come-first-served and nearest-location-first dispatch?**

Proposed Solution/ Innovation
Briefly describe your proposed solution and explain how it addresses the identified problem (Max 300 words)

**AquaReach is a web-based decision-support platform that helps emergency water dispatchers decide which communities to serve, how much water to send, in what order, and where residents need extra help.**

**For each service location, such as a residential block, housing area or student hostel, AquaReach combines population, usable stored water, essential demand, receiving-tank capacity, road access and the share of residents who need help collecting water. Data can be entered by the utility or submitted by building managers and residents' committees through a simple "tank check-in" form, so no new sensors are required.**

**AquaReach estimates a range for the time remaining before each community runs out of water. A prioritisation and optimisation engine then recommends delivery allocations and sequences that respect tanker capacity, available water, travel and refilling time, and receiving-site capacity, with the goal of minimising unmet essential demand while avoiding extreme waits for any single community.**

**A map-based dashboard explains every recommendation in plain language (for example, "Block C: about 3 hours of water left, 20% of residents need assistance"), flags missing or outdated data and lets dispatchers override the plan. Assistance flags alert community volunteers to carry water the "last 50 metres" to residents who cannot collect it themselves. All dispatch decisions remain under human control.**

**We have built a prototype simulation and tested it on 500 synthetic disruption scenarios (8 sites, 3 tankers, 48 hours). AquaReach reduced unmet essential water by 63% compared with first-come-first-served and 80% compared with nearest-location-first, and cut the longest time any site went without water to 0.8 hours (from 2.6 and 19.3 hours), at the cost of 11–16% more travel. These results are simulated and will next be validated with real operational data.**

## What Makes Your Solution Innovative?

**AquaReach brings together three things that are usually handled separately: the time until each community's water runs out, fairness-aware allocation and practical access support.**

**Rather than claiming a new routing algorithm, it combines established methods into a transparent workflow that dispatchers can trust and question. Every recommendation shows why a site comes first and what the trade-off is, such as slightly longer travel in exchange for fewer people left without water. Scenario comparisons show how alternative decisions would change waiting times and unmet demand.**

**AquaReach also recognises that water arriving at a tank is not the same as residents being able to use it, and uses assistance flags to close that "last 50 metres" gap. This turns the human right to water into a measurable operational goal. Its value is already demonstrated through controlled simulation against common dispatch rules.**

Target Users / Beneficiaries
Who would potentially use or benefits from your solution?

**Primary users are water utility emergency response teams and dispatchers responsible for coordinating alternative water supplies. Secondary users include local authorities, residential building managers and community organisations assisting with distribution.**

**Beneficiaries are residents affected by water disruptions, particularly communities with limited water storage and people who face barriers to collecting water, including older residents and persons with disabilities. The initial pilot will focus on a small group of residential communities or university accommodation sites, subject to stakeholder participation and data availability.**

## SECTION 4 : IMPACT & IMPLEMENTATION

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Expected mpact

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- [ ] Water Conservation
- [ ] Reduced Water Lossess
- [x] Improved Water Supply Management
- [ ] Flood Resilience
- [x] Climate Adaptation
- [ ] River/ Catchment Protection
- [ ] Waste Reduction/ Resource Recovery
- [ ] Improved Environmental Awareness
- [x] Improved Environmental Management
- [x] Economic Benefits
- [x] Social Benefits
- [ ] Other

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Potential for Implementation

A working simulation prototype and an interactive dispatch dashboard have already been built and tested against two baseline dispatch rules. AquaReach can be deployed as a lightweight web application using CSV uploads and open-source optimisation libraries, so it needs no new sensors or infrastructure. Implementation will begin with a shadow-mode pilot at six to ten sites, such as university hostels or high-rise residential blocks, where AquaReach generates recommendations alongside real dispatch decisions without controlling them. Partners could include water utilities, local councils, building managers and residents' associations. All dispatch decisions remain under human control, and only aggregated community data is used. After a successful pilot, the system can be integrated with existing emergency workflows and scaled to whole districts, other states and other emergency supplies such as flood relief.